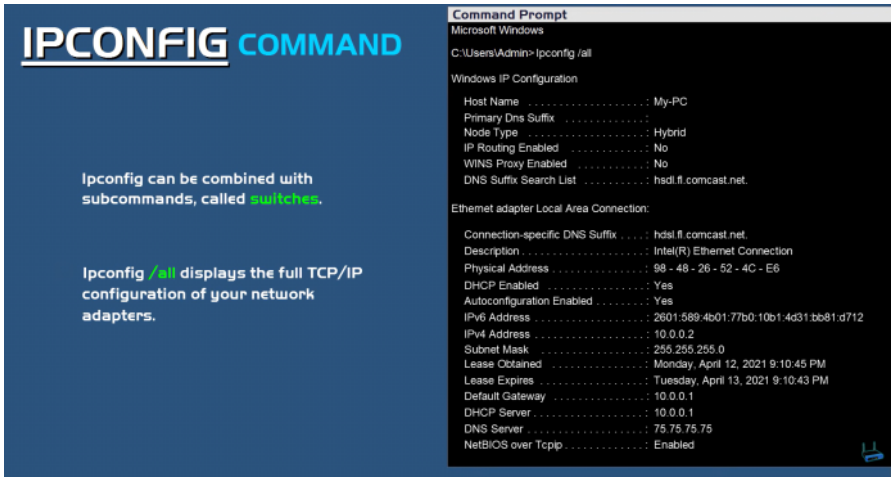
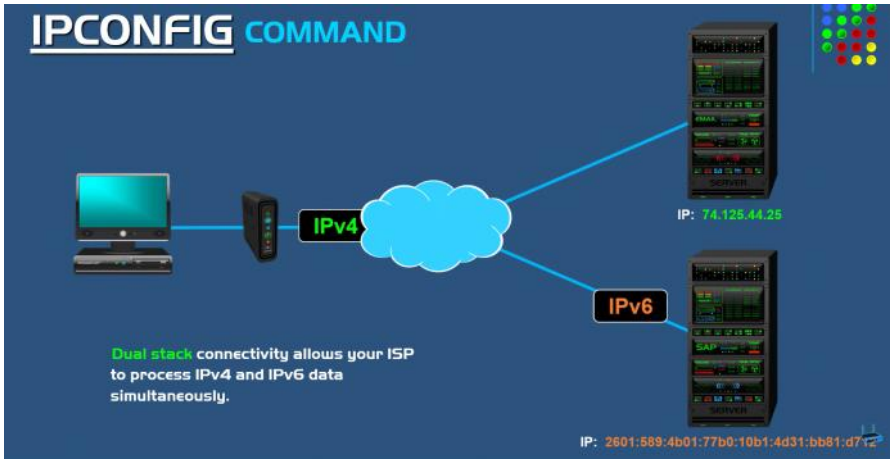
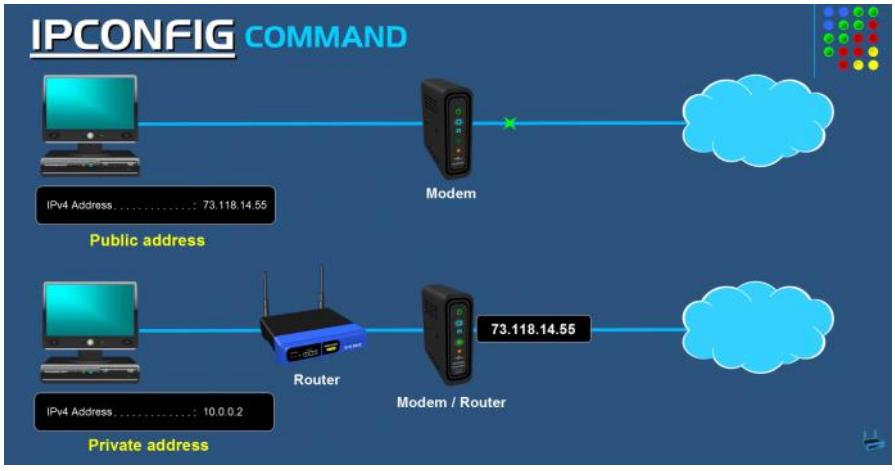
IPCONFIG

Saturday, April 18, 2026 1:43 PM

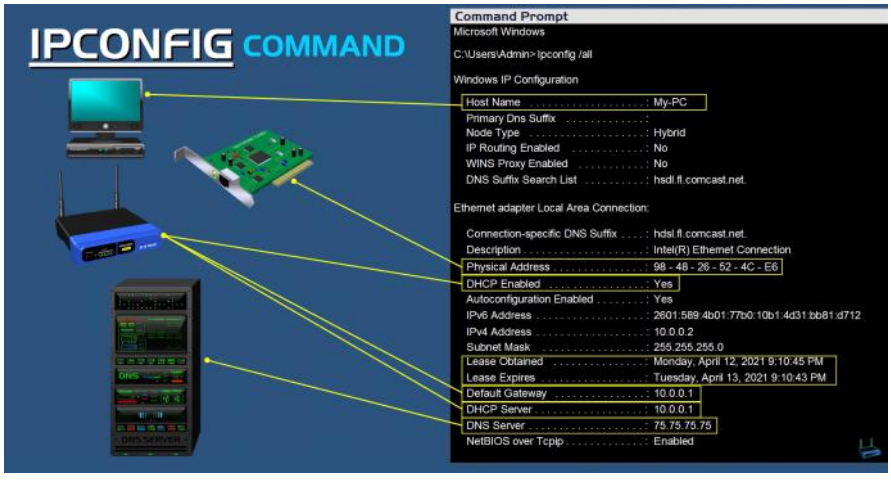
IPconfig is a command-line tool for Windows computers used to display TCP/IP network configuration details, such as IP addresses, subnet masks, and default gateways. It is a vital utility for network troubleshooting.







Key Concepts and Functions:



- Dual Stack: Computers often use both IPv4 and IPv6 addresses simultaneously. This setup is known as a dual stack, implemented to support the transition from the older IPv4 protocol to the modern IPv6 standard, ensuring compatibility across the internet.
- Command Switches:
- ipconfig /all: Displays a comprehensive report of network adapter settings, including the host name, physical (MAC) address, DHCP server information, and DNS server addresses.
- ipconfig /displaydns: Shows the contents of the local DNS resolver cache, which stores previously resolved domain names and their corresponding IP addresses to speed up web browsing.
- ipconfig /flushdns: Clears the DNS cache. This is useful for resolving connectivity issues caused by outdated records, preventing DNS spoofing attacks, or hiding website search behavior.
- ipconfig /?: Provides a complete list of all available command switches.